CSW-332 MACHINE LEARNING AND APPLICATIONS

Time Allotted = 90 min.

Answer all questions –

1. Write a Python program for the Random Forest (RF) supervised machine learning classifier using the Iris dataset. The program should demonstrate the following operations:
    a. Load the Iris Dataset and split it into training and testing samples.
    b. Train the RF ML classifier.
    c. Evaluate the model using performance metrix derived from Confusion Matrix.
    d. Validate the classifier using multiple test samples.

2. Write a Python program for the Support Vector Machine (SVM) supervised machine learning classifier using the Iris dataset. The program should demonstrate the following operations:
    a. Load the Iris Dataset and split it into training and testing samples.
    b. Perform feature scaling.
    c. Train the SVM ML classifier.
    d. Evaluate the model using performance metrix derived from Confusion Matrix.
    e. Validate the classifier using multiple test samples.

3. Write a Python program for Convolutional Neural Network (CNN) in Machine Learning using the MNIST dataset. The program should demonstrate the following operations:
    a. Load the necessary Libraries.
    b. Load the MNIST dataset and split it into training and testing samples.
    c. Build and train the CNN model.
    d. Evaluate the model performance.
    e. Predict the output for any one sample.